

1.10.25

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Question: If a line makes angles 90° , 60° , and 30° with the positive directions of the X , Y , and Z axes respectively, find its direction cosines.

Solution:

Definition of Direction Cosines: The direction cosines of a vector are the cosines of the angles that the vector makes with the positive x -axis, y -axis, and z -axis in $3D$ space. These direction cosines give us a way to describe the orientation of a vector in space

Angle (α)	$\cos(\alpha)$	Value	Axis
90°	$\cos(90^\circ) = 0$	$l = 0$	x-axis
60°	$\cos(60^\circ) = \frac{1}{2}$	$m = \frac{1}{2}$	y-axis
30°	$\cos(30^\circ) = \frac{\sqrt{3}}{2}$	$n = \frac{\sqrt{3}}{2}$	z-axis

TABLE 0: Variables Used

Let the direction cosines be l, m, n , which are the cosines of the angles that the line makes with the X , Y , and Z axes respectively.

$$l = \cos(90^\circ) = 0 \quad (0.1)$$

$$m = \cos(60^\circ) = \frac{1}{2} \quad (0.2)$$

$$n = \cos(30^\circ) = \frac{\sqrt{3}}{2} \quad (0.3)$$

A key property is that the sum of the squares of the direction cosines equals 1:

$$l^2 + m^2 + n^2 = 1$$

$$\text{Direction cosines of vector } x = \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \quad (0.4)$$

Refer to Figure

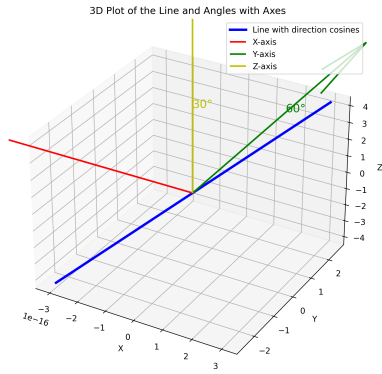


Fig. 0.1